

MATPREDICT · MATING-TYPE REPORT

mucoro_absidia_cylindrospora

taxid 213030 · Mucoromycota

Run by MATPredict 0.6.1 · database 2b90ce832408 · NCBI E-utilities · rendered 2026-10-08 04:23 UTC by MATPredict 0.6.1

Result

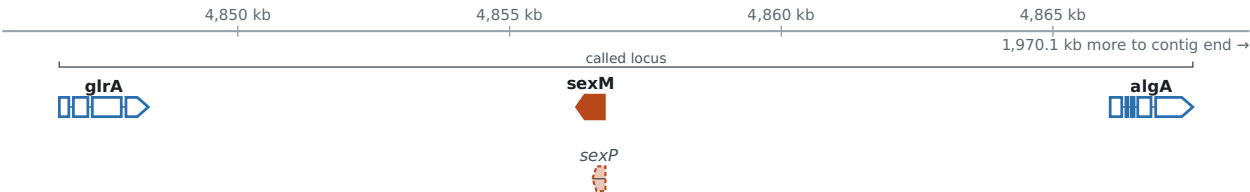
MATING TYPE

Minus medium confidence

- **Minus**: complete locus, medium confidence, 20.9 kb on CDRYNH010000023.1. Gene order: glrA · sexM · sexP · algA.

Called loci 1

Minus MAT family · Mucoromycota medium confidence



Gene order on CDRYNH010000023.1.

core MAT gene variable flank gene model search hit only

LOCATION	LOCUS	GENES MODELLED
CDRYNH010000023.1:4,846,711–4,867,574 · 20.9 kb	Complete locus: MAT gene(s) with their flanking genes.	3 of 4 gene hits
NEAREST CONTIG END	REGION (SAMTOOLS)	
2.0 Mb	CDRYNH010000023.1:4846711-4867574	

Idiomorph scores**Minus** 110.7 **Plus** 60.0Margin over the next idiomorph: **50.7**

Score: HMM bitscore of the MAT protein against each idiomorph's profile.

Profile-HMM classifier**Minus** 110.7 **Plus** 60.0Margin **50.7** (a call needs 25.0 or more). Scored: sexM, sexM (model).**Overlapping hits resolved**

Where genes of two idiomorphs hit the same place, the stronger hit was kept.

kept	identity %	set aside	identity %	overlap %	basis
sexM	43.2	sexP	37.5	95.0	higher identity

GENE EVIDENCE

Gene	Role	Model	Identity %	E-value	Position (strand)	Exons	Reference record
 glrA	variable flank	gene model exonerate	78.2	< 1e-180 694 bits	4,846,714–4,848,351 +	4	86630_cbs357-93_ MAT_Minus
 sexM	core MAT	gene model exonerate	38.5	5.5e-36 135 bits	4,856,220–4,856,753 -	1	3231046_uomau377_ MAT_Minus
 sexP	core MAT	hit only tblastn	37.5 cov 22.0%	8.7e-08 54 bits	4,856,526–4,856,765 -	-	36080_nrnl-3631_ MAT_Plus
 algA	variable flank	gene model exonerate 4 frameshifts	63.5	1.1e-106 269 bits	4,866,054–4,867,574 +	5	3231046_uomd23-19_ MAT_Plus
tptA	<i>expected, not found</i>						
rnhA	<i>expected, not found</i>						

What was searched

ROUTING

Phylum given. The phylum was given when the run was started; taxonomy was not used.

FAMILIES SEARCHED

Mucoromycota:MAT

GENETIC CODE

1

TAXONOMY

NCBI E-utilities

Withheld candidate loci 2

Places with some MAT-gene evidence that did not pass a quality gate. Most genomes have several. They are listed so a known locus can be checked against them, not because they are likely calls.

Reasons: no MAT gene passed the MAT-gene check (1); too few of the family's genes found (1).

WITHHELD LOCI (2)

family	location	idiomorph	genes	modelled	reason
MAT	CDRYNH010000012.1: 717,983–761,620	undetermined	sexP, sexM, rnhA, glrA	2	no MAT gene passed the MAT-gene check
MAT	CDRYNH010000012.1: 6,447,171–6,496,043	undetermined	sexM, rnhA	1	too few of the family's genes found

Provenance

GENOME FILE

mucoro_absidia_
cylindrospora.fna

ASSEMBLY

34 contigs · 53.7 Mb · N50 6.4
Mb

GENOME SHA-256

2d28797d9e36bcad7433d0b81f9
f7ff2cdcd455c275c40353a1856
99f030661e

MATPREDICT

0.6.1

DATABASE

2b90ce832408b109 · 163
records

TAXONOMY

NCBI E-utilities

PARAMETERS

phylum=Mucoromycota,
min_hits=2,
require_core_role=True,
max_polished_clusters_per_family=6,
polish_strong_identity=50.0

RUN STARTED

2026-10-08T04:22:53Z

RUN TIME

53 s

FILES

detection_report.yaml, detected_loci.gff3

Cite MATPredict as given in CITATION.cff at github.com/stajichlab/MATPredict. The reference records named in the gene tables are curated database entries; each record's metadata cites its source publication.

How to read this report

Idiomorph

The form of the mating-type locus a strain carries (for example MAT1-1 or MAT1-2, Plus or Minus).

Basidiomycota HD and PR alleles are named after the reference they match.

Low confidence

A single isolated gene hit, or no core MAT gene.

High confidence

Every expected core MAT gene was found and modelled, with a conserved flanking gene where the family has one, on one contig.

Model status

Gene model: the reference protein was aligned to the genome to build the gene (exonerate, miniprot).

Models differ: the two tools disagree on the structure; the alternate model is shown. Hit only: a search hit with no model, weaker evidence.

Medium confidence

The core genes were found, but a gene could not be modelled, no conserved flank was found, or the locus is split (one step down).

Identity

Percent amino-acid identity between the gene (or hit) and the closest reference protein.

Reference record

The curated database record whose protein matched best. Each record's metadata cites the publication or deposit it comes from.

LIMITS

- MATPredict reads an assembly; it is not a mating assay. Two idiomorphs in one assembly can come from a homothallic species, a mixed culture or a merged diploid assembly.
- Fragmented assemblies can split a locus or lose its flanking genes; an uncalled genome is weak evidence of absence.
- Lineages with no curated record are searched with the whole phylum's references; family labels there are less reliable.